

GrounUp Enterprise Specification

Phase 04 - Master Library Registry

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1. Executive Summary

Phase 04 defines the governed Master Library Registry and the single source of truth for GrounUp master data across SaaS, web, mobile, white-label and OEM deployments.

2. Master Library Catalog

ID	Library	Domain	Scope
LIB-001	Master Service Library	Estimating	Global
LIB-002	Master Resource Library	Estimating	Global
LIB-003	Master Material Library	Cost	Global
LIB-004	Master Equipment Library	Fleet	Global
LIB-005	Master Fleet Library	Fleet	Tenant
LIB-006	Master Labor Library	HRM	Tenant
LIB-007	Master Crew Library	Operations	Tenant
LIB-008	Master Vendor Library	Procurement	Tenant
LIB-009	Master Customer Library	CRM	Tenant
LIB-010	Master Subcontractor Library	Procurement	Tenant
LIB-011	Master Production Rate Library	Estimating	Hybrid
LIB-012	Master Cost Code Library	Finance	Hybrid
LIB-013	Master Assembly Library	Estimating	Hybrid
LIB-014	Master CSI Library	Standards	Global
LIB-015	Master DOT Bid Item Library	Infrastructure	Global
LIB-016	Master Municipal Bid Item Library	Infrastructure	Global
LIB-017	Master Unit Library	Reference	Global
LIB-018	Master Currency Library	Reference	Global
LIB-019	Master Tax Library	Finance	Tenant
LIB-020	Master Safety Library	Safety	Hybrid
LIB-021	Master Specification Library	Documents	Hybrid
LIB-022	Master Drawing Symbol Library	Takeoff	Global
LIB-023	Master AI Knowledge Library	AI	Hybrid
LIB-024	Master Prompt Library	AI	Hybrid
LIB-025	Master Risk Library	Risk	Hybrid
LIB-026	Master Quality Library	Quality	Hybrid
LIB-027	Master Inspection Library	Quality	Hybrid
LIB-028	Master Checklist Library	Operations	Hybrid
LIB-029	Master Workflow Library	Platform	Hybrid
LIB-030	Master Report Library	Analytics	Hybrid
LIB-031	Master Dashboard Library	Analytics	Hybrid
LIB-032	Master Notification Library	Platform	Hybrid
LIB-033	Master Template Library	Documents	Hybrid

ID	Library	Domain	Scope
LIB-034	Master Document Library	Documents	Hybrid
LIB-035	Master Formula Library	Calculation	Global
LIB-036	Master Location Factor Library	Cost	Global
LIB-037	Master Regional Pricing Library	Cost	Hybrid
LIB-038	Master Equipment Productivity Library	Production	Global
LIB-039	Master Crew Productivity Library	Production	Hybrid
LIB-040	Master Disposal Library	Hauling	Tenant
LIB-041	Master Trucking Library	Hauling	Hybrid
LIB-042	Master Rental Rate Library	Cost	Tenant
LIB-043	Master Fuel Library	Cost	Hybrid
LIB-044	Master Maintenance Library	Fleet	Tenant
LIB-045	Master Asset Library	Fleet	Tenant
LIB-046	Master Inventory Library	Inventory	Tenant
LIB-047	Master Time Code Library	HRM	Tenant
LIB-048	Master Payroll Code Library	HRM	Tenant
LIB-049	Master Accounting Mapping Library	Finance	Tenant
LIB-050	Master AI Training Library	AI	Hybrid

3. Canonical Fields

Field	Name	Type	Req.	Definition
FLD-001	library_item_id	string	Yes	Immutable globally unique identifier
FLD-002	tenant_id	uuid	Conditional	Owning tenant; null for global records
FLD-003	library_id	string	Yes	Parent library identifier
FLD-004	item_code	string	Yes	Human-readable unique code within scope
FLD-005	name	string	Yes	Primary display name
FLD-006	short_name	string	No	Compact display label
FLD-007	description	text	Yes	Controlled definition
FLD-008	category_id	string	Yes	Primary category reference
FLD-009	subcategory_id	string	No	Secondary category reference
FLD-010	status	enum	Yes	Draft, Review, Approved, Active, Deprecated, Archived
FLD-011	version	semver	Yes	Semantic version
FLD-012	effective_from	date	Yes	Start of validity
FLD-013	effective_to	date	No	End of validity
FLD-014	unit_id	string	Conditional	Primary unit reference
FLD-015	source_type	enum	Yes	Internal, vendor, standard, government, historical, AI-derived
FLD-016	source_reference	string	Yes	Traceable source citation
FLD-017	confidence_score	decimal	Yes	0.00 to 1.00 confidence
FLD-018	approval_status	enum	Yes	Pending, Approved, Rejected, Superseded
FLD-019	approved_by	uuid	Conditional	Approver identity
FLD-020	approved_at	datetime	Conditional	Approval timestamp
FLD-021	created_by	uuid	Yes	Creator identity
FLD-022	created_at	datetime	Yes	Creation timestamp
FLD-023	updated_by	uuid	Yes	Last editor identity
FLD-024	updated_at	datetime	Yes	Last update timestamp
FLD-025	tags	array	No	Search and AI tags
FLD-026	keywords	array	No	Synonyms and search keywords
FLD-027	relationships	array	No	Typed references to related records
FLD-028	custom_fields	json	No	Tenant-configured extensions
FLD-029	white_label_visibility	enum	Yes	Platform, tenant, reseller, hidden
FLD-030	audit_hash	string	Yes	Integrity hash of approved payload

4. Governance and Rules

BR-001 - Single Source of Truth

Every governed master value must originate from one registered library record.

BR-002 - Immutable Identity

Approved records retain immutable IDs across revisions and deployments.

BR-003 - Tenant Isolation

Tenant-owned records must never be visible across tenants without explicit sharing.

BR-004 - Global Inheritance

Tenants may inherit global records and override only fields marked configurable.

BR-005 - Approval Before Use

Draft or rejected records cannot be used in production calculations.

BR-006 - Effective Dating

Only records valid for the transaction date may be selected.

BR-007 - Version Pinning

Estimates and projects pin the exact library version used.

BR-008 - No Silent Replacement

Superseding a record must not silently rewrite historical transactions.

BR-009 - Source Provenance

Every cost, rate, formula and AI knowledge record requires a traceable source.

BR-010 - Confidence Governance

AI-derived or inferred records require confidence scoring and review thresholds.

BR-011 - Unit Compatibility

Relationships and formulas may combine only dimensionally compatible units.

BR-012 - White Label Segregation

Brand-specific templates and content must remain scoped to the owning brand.

BR-013 - Custom Field Safety

Tenant custom fields may extend but not redefine protected canonical fields.

BR-014 - Deprecation Window

Deprecated items remain searchable for historical traceability but cannot be newly selected.

BR-015 - Four-Eyes Approval

High-impact pricing, formula and compliance changes require separate author and approver.

BR-016 - Bulk Import Staging

Bulk imports enter staging and require validation before publication.

BR-017 - Duplicate Prevention

Potential duplicates must be flagged before approval.

BR-018 - Regional Precedence

Project-specific pricing overrides regional, tenant and global defaults in controlled order.

BR-019 - Audit Completeness

Every create, edit, approval, rejection and archive action must be auditable.

BR-020 - Rollback Capability

Published versions must support rollback without deleting audit history.

5. Workflows and Engines

WF-001 - Create Library Item: Draft -> Validate -> Review -> Approve -> Publish

WF-002 - Revise Library Item: Clone Active -> Edit -> Compare -> Approve -> Supersede

WF-003 - Bulk Import: Upload -> Stage -> Map -> Validate -> Resolve -> Approve -> Publish

WF-004 - Tenant Override: Inherit -> Override Allowed Fields -> Validate -> Activate

WF-005 - Deprecation: Impact Analysis -> Notice -> Replacement Mapping -> Deprecate

WF-006 - Emergency Correction: Flag Critical -> Dual Approval -> Hotfix Version -> Notify

WF-007 - Duplicate Resolution: Detect -> Compare -> Merge/Relate/Reject -> Audit

WF-008 - White Label Publication: Brand Scope -> Preview -> Approve -> Publish

WF-009 - AI Knowledge Intake: Ingest -> Chunk -> Tag -> Evaluate -> Approve -> Index

WF-010 - Periodic Review: Schedule -> Owner Review -> Revalidate -> Renew/Revise/Archive

ENG-001 - Library Registry Engine: Registers library definitions, ownership and lifecycle.

ENG-002 - Inheritance and Override Engine: Resolves global, reseller, tenant and project precedence.

ENG-003 - Version and Effective-Date Engine: Selects valid immutable versions by context date.

ENG-004 - Validation Engine: Executes schema, business and relationship validations.

ENG-005 - Duplicate Detection Engine: Uses code, name, attributes and semantic similarity.

ENG-006 - Relationship Graph Engine: Maintains typed dependencies and impact analysis.

ENG-007 - Approval and Publication Engine: Controls review, approval, publish and rollback.

ENG-008 - Search and Indexing Engine: Provides faceted, full-text and semantic retrieval.

ENG-009 - Import and Mapping Engine: Stages, maps and validates CSV/Excel/API imports.

ENG-010 - White Label Resolution Engine: Applies brand-specific visibility, terminology and templates.

ENG-011 - Confidence and Provenance Engine: Scores reliability and enforces source evidence.

ENG-012 - Audit Integrity Engine: Creates tamper-evident hashes and event history.

6. API, UX, AI and Acceptance

The API includes 24 governed endpoints. The user experience provides role-based library administration, cross-library search, import wizards, approval queues, version comparison, dependency impact and white-label preview. AI cannot self-approve records. Acceptance is based on 97 requirements and 97 mapped test cases.